```
Instance variable
class Man
{
        private String name;
        private int age;

         public void setName(String name)
         {
                  this.name = name;
         }

         public String getName()
         {
              return this.name;
         }

        public void setAge(int age)
         {

           if (age > 0){

              this.age = age;

              }
        }

        public int getAge()
         {

            return this.age;
        }

     public void speak()
     {

          System.out.println(getName() + " is speaking
now");
     }

}
```

```
public class A_InstanceVariableMain {
     public static void main(String[] args){

          Man man;

          man = new Man();

          man.setName("RamKumar");

          System.out.println("man's name is " +
man.getName());
          man.speak();
     }
}
```

Local variable

```
class B_LocalVariableMain
{
     int data=50;
     void print()
     {

          //int number=90;
         int number;
     //   System.out.println(number);

     }

     public static void main(String args[])
     {

          B_LocalVariableMain  object  =  new
B_LocalVariableMain();
          System.out.println(object.data);
          object.print();
```

```
    }
}
```

Static variable

```
//static  variable

class Pen
{
  static int count=0;
  // static int count;
  // count = 0; //?
  String name;
}

class C_StaticVariableMain
{

  public static void main(String args[])
  {

     Pen.count++;//1

    Pen.count++;//2

    System.out.println("No of objects  " + Pen.count);


  }
}
```

Static method

```
class MathUtility
{
    public static int findSquare(int number) {
        return number * number;
    }
```

```
    public static int findCube(int number) {
        return number * number * number;
    }
}

public class D_StaticMethod {

      public static void main(String[] args) {

             System.out.println("Square : " +
MathUtility.findSquare(5));
             System.out.println("Cube   : " +
MathUtility.findCube(5));
         }

}
```

When static method

```
class Calc {
    int age; //instance variable

    //if method is not using object data
    public static int add(int num1, int num2) {
     //System.out.println(age);
        return num1 + num2;
    }
}


class F_WhenStaticMethod {

     public static void main(String[] args)
     {
                int sum = Calc.add(10, 20);
                System.out.println(sum);
    }

}
```

When instance method

```
class Person {

    int id;
    String name;
//if method is  using object data
    public void display() {
        System.out.println(id + " " + name);
    }
}
class E_WhenInstanceMethod
{
    public static void main(String[] args) {

     Person person = new Person();
     person.id = 101;
     person.name = "Abi";

     person.display();
    }
}
```

Garbage collection

```
class Student {

    int id = 101;
}

public class A_GC {

    public static void createStudent()
    {

        Student student = new Student();

        System.out.println("Student Object Created");
```

```
    }

    public static void main(String[] args) {

        createStudent();

       // System.out.println(student.id); // Compilation
Error
    }
}
```

Garbage collection

```
class Employee {

    int id = 101;
}

public class B_GC {

    public static void createStudent() {

     Employee employee = new Employee();

        System.out.println("Student Object Created");

        employee = null;

        System.out.println("Reference variable assigned to
null");
    }

    public static void main(String[] args) {

        createStudent();

        // System.out.println(student.id); // Compilation
Error
    }
}
```